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From statistics to (hyper)graphs.

On the benefit of higher-order interactions for a good hierarchical structure

Presentation at the NEO-AMELEAS Workshop

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Outline

I) Clustering: From Statistics to (Hyper)graphs

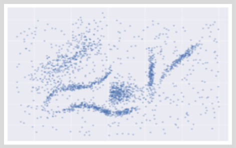
II) Higher-Order DELAUNAY Mosaic: The Object Containing Hierarchical Information

III) A Few Examples in 3D

01
Clustering: From Statistics to (Hyper)graphs

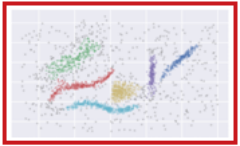
Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$

Hartigan's Model [5]: **High-density clusters**



Raw data $\mathcal{X}$

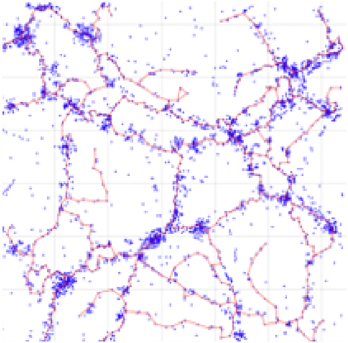
HDBSCAN[1] [2, 9]



High-density clusters

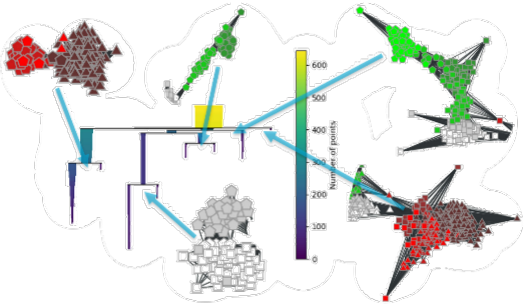
[1] Hierarchical Density-Based Spatial Clustering of Applications with Noise

Some Examples



3D Cosmology: cosmic filaments [6]

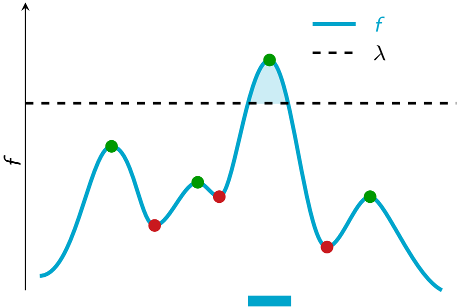
Some Examples



Italian olive oil classification [14] ($\mathcal{X} \subset \mathbb{R}^8$): hierarchical result [7, 8]

Hartigan's Model: High-Density Cluster Hierarchy

Level set $\{f \geq \lambda\}$



Hierarchical tree

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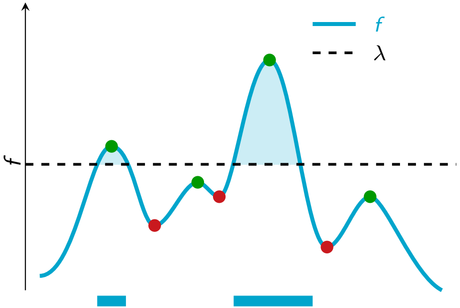
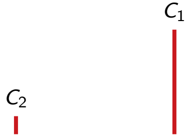


Hierarchical tree



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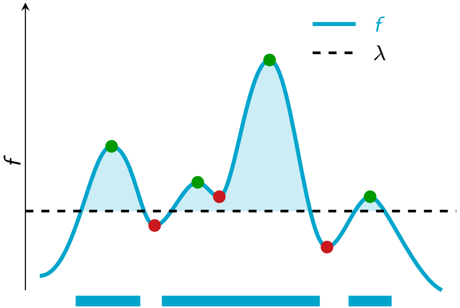
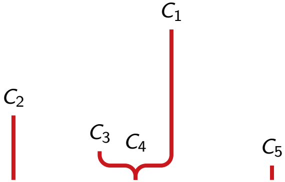


Hierarchical tree



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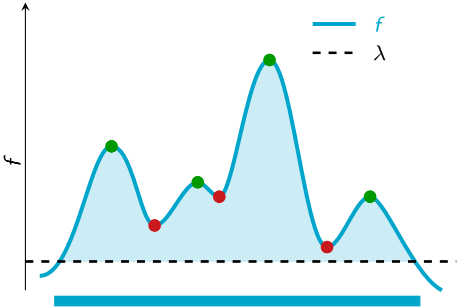
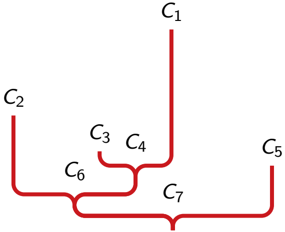


Hierarchical tree



Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: Estimation

f unknown → **Estimation $\widehat{f}$**

Expensive naive methods
(discretization of $\mathbb{R}^d$)

Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: Density Estimator

$$\widehat{f} \leftarrow \widehat{f}_{K-\mathrm{NN}}$$

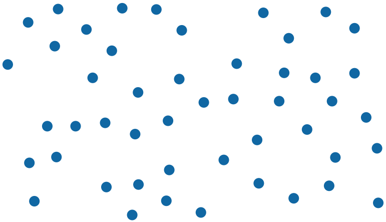
Estimator: K-Nearest Neighbors (K-NN)

$$\forall \boldsymbol{x} \in \mathbb{R}^{\boldsymbol{d}}, \quad \widehat{\boldsymbol{f}}_{\boldsymbol{K}-\mathrm{NN}}(\boldsymbol{x}) = \frac{\boldsymbol{K}}{\boldsymbol{n}\,\boldsymbol{\omega_d}\,\boldsymbol{r_K}(\boldsymbol{x})^{\boldsymbol{d}}}$$

$$\boldsymbol{r_K}(\boldsymbol{x}) = \min\{\boldsymbol{r} \geq 0 \mid |\boldsymbol{B}(\boldsymbol{x},\boldsymbol{r}) \cap \mathcal{X}| \geq \boldsymbol{K}\}$$

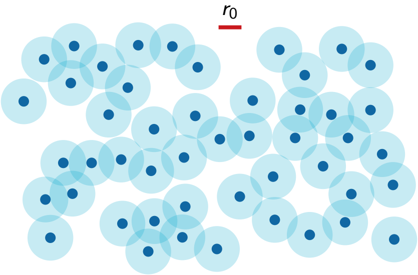
- n: number of points in $\mathcal{X}$
- ω_d: volume of the unit ball in $\mathbb{R}^d$
- $r_K(x)$: distance to the K-th nearest neighbor of x
- $\omega_d\, r_K(x)^d$: volume of the ball containing the K neighbors

Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: The Case $K = 1$



Poisson point cloud $\mathcal{X} \subset \mathbb{R}^2$

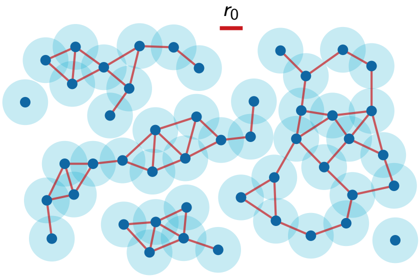
Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: The Case $K = 1$



$$\widehat{\lambda}(x) \geq \lambda_0$$
$$\Leftrightarrow r_1(x) \leq r_0$$

High-density level sets
$=$ **Boolean model**

Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: The Case $K = 1$

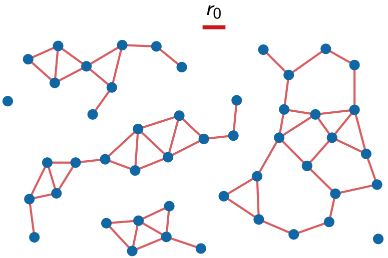


$$\widehat{\lambda}(x) \geq \lambda_0$$
$$\Leftrightarrow r_1(x) \leq r_0$$

High-density clusters
$=$ **Connected components** of
$G(\mathcal{X}, r_0)$

Geometric graph [12]

Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: The Case $K = 1$



$$\boldsymbol{r_1(x) \leq r_0}$$

High-density clusters
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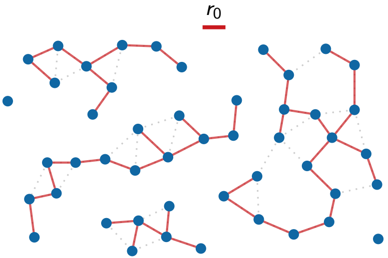
Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: The Case $K = 1$

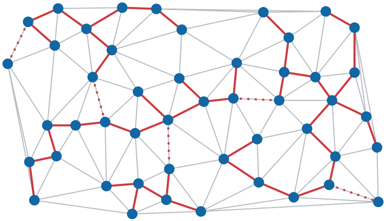


$$\boldsymbol{r_1(x) \leq r_0}$$

High-density clusters
$=$ **Connected components** of
$\mathrm{MST}_{r_0}(\mathcal{X})$

Pruned Minimum Spanning Tree

The Minimum Spanning Tree and the Delaunay Triangulation



The Euclidean Minimum Spanning Tree is a subgraph of the Delaunay triangulation [1]:

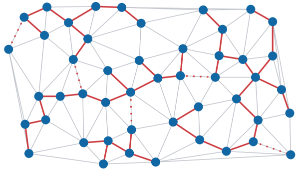
$$\mathrm{MST} \subseteq \textsc{Delaunay}.$$

In the plane $\mathbb{R}^2$:

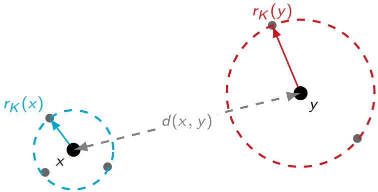
- The Delaunay triangulation has at most $3n - 6$ edges (linear complexity).
- It can be computed in $\mathcal{O}(n \log n)$.

Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: High-Density Clusters

- If $\widehat{f} = \widehat{f}_{1-\mathbf{NN}}$: density estimator by 1-nearest neighbor.
- Construction of the **Minimum Spanning Tree** (MST):
  - $\mathcal{O}(n^2)$ in the worst case.
  - Geometric optimizations: $\mathcal{O}(n \log n)$ in $\mathbb{R}^2$.
- Clustering algorithms:
  - **Single-Linkage** $\longrightarrow$ **HDBSCAN**



From Single-Linkage to Robust Single-Linkage



$$d_{\mathrm{mreach},K}(x, y) = \max\{r_K(x), r_K(y), d(x, y)\}$$

Chaining effect: Classic Single-Linkage (1-NN) is highly sensitive to noise that artificially connects two clusters.

Robust Single-Linkage [3]:

- Uses the **core distance** $r_K(x)$ (radius of the K-NN) as a local density estimator.
- Replaces the geometric distance with the **mutual reachability distance** $d_{\mathrm{mreach},K}$.
- Pushes isolated noise points away by artificially increasing their distance to other points.

From Robust Single-Linkage to (H)DBSCAN

Limitation of Robust Single-Linkage:

- Overly restrictive for peripheral points.
- Discards border points without K neighbors, reducing recall by misclassifying them as noise.

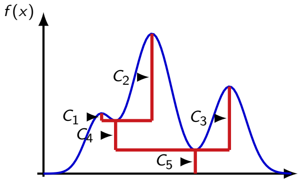
DBSCAN [4] (Ester *et al.*, 1996):

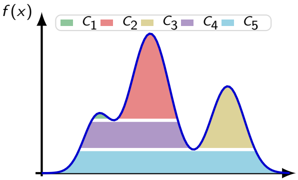
- Distinguishes **core points** ($r_K(x) \leq r$) from **border points** (reachable from a core but lacking K neighbors).
- Improves classification of cluster boundaries.

From DBSCAN to HDBSCAN

DBSCAN requires a single global threshold r. **HDBSCAN** [2] (Campello *et al.*, 2013) varies this threshold continuously to eliminate r and construct a complete hierarchy.

From DBSCAN to HDBSCAN: Excess of Mass



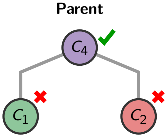


Excess of mass [2]:

- Stability of a cluster C at λ:

$$E(C) = \int_C \big(f(x) - \lambda\big)\,\mathrm{d}x$$

- Corresponds to the **colored areas**.

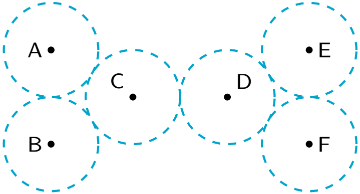


Violet Area $>$ **Green Area** $+$ **Red Area**
$\implies$ Keep C_4

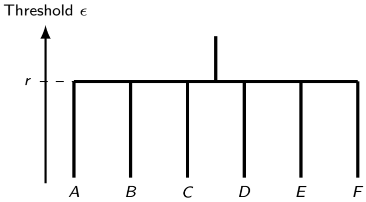
(vs. splitting into C_1 and C_2)

Robust SL / (H)DBSCAN ($K = 2$)

Point cloud and balls of radius r

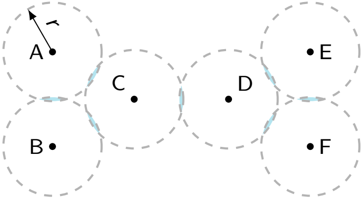


Hierarchical tree

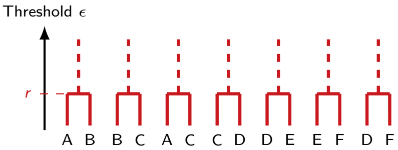


The True 2-NN Hierarchy: Scale r

Point cloud and balls at level r



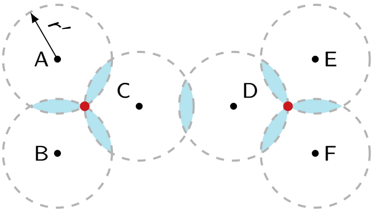
Hierarchical tree



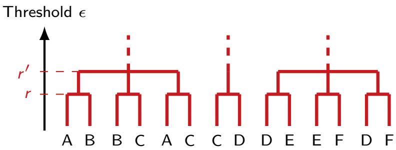
Step 1: The 7 segments merge at threshold r.

The True 2-NN Hierarchy: Scale r'

Triangles and 3-intersections at r'



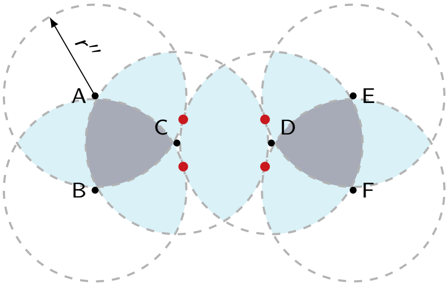
Hierarchical tree



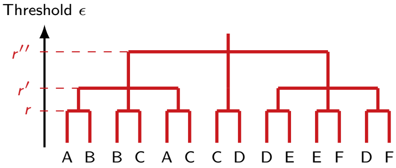
Step 2: Merging of triangles at r' via 3-intersections. *CD* remains isolated.

The True 2-NN Hierarchy: Scale r''

Global connectivity at r''



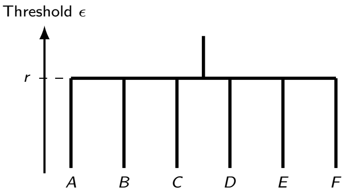
Hierarchical tree



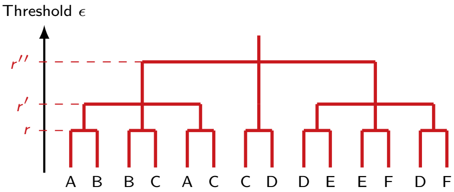
Step 3: Global connectivity at r'' through tangency points.

Robust SL vs. True 2-NN Hierarchy: Comparison

Robust SL Hierarchy ($K = 2$)



True 2-NN Hierarchy



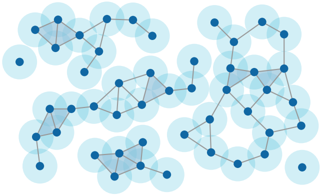
Summary: The true 2-NN hierarchy isolates the connection bridge C-D, unlike Robust SL.

Clustering on Euclidean Data $\mathcal{X} \subset \mathbb{R}^d$: High-Density Clusters ($K \geq 2$)

- If $\widehat{f} = \widehat{f}_{K-\mathbf{NN}}$ ($K \geq 2$): density estimator by K-nearest neighbors.

- Transition to **hypergraphs** (geometric complexes):
  - Edges and triangles appear at the intersections of balls of radius r.

- Higher-order clustering algorithms:
  - Connected components are replaced by K-**polyhedra**.
  - For $K \geq 2$, clusters (polyhedra) can overlap.



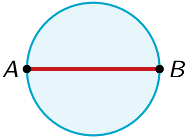
02

Higher-Order Delaunay Mosaic: The Object Containing Hierarchical Information

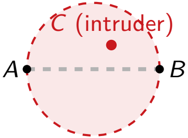
The Case $K = 1$: The Gabriel Graph

- **Definition**:
  - An edge $[A, B]$ is a Gabriel edge if its **diametral disk** does not contain any other point in $\mathcal{X}$.
- **Intermediate structure**:
  - Key structure between the MST and the triangulation:

$$\text{MST} \subseteq \textsc{Gabriel}\ \text{Graph} \subseteq \textsc{Delaunay}$$



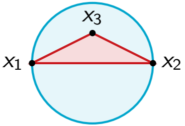
Gabriel edge (empty)



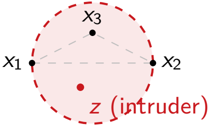
Non-Gabriel (non-empty)

Generalization to Order K: Splitting Simplices

- **K-splitting simplex**:
  - It is a K-simplex that has a **real influence** (causes clusters to merge) on the K-NN hierarchy.
  - **Case** $K = 1$: 1-splitting simplices are precisely the **edges of the MST**.
- **Gabriel condition**:
  - **Theorem**: any K-splitting simplex is necessarily a **Gabriel simplex** ($\mathring{B}_\sigma \cap (\mathcal{X} \setminus \sigma) = \varnothing$).
  - **Computational domain**: the **higher-order Delaunay mosaic** [11].



Gabriel triangle (empty)



Non-Gabriel (non-empty)

The K-Minimum Spanning Tree Analogy

Similar to $K = 1$, the higher-order hierarchy relies on a sparse combinatorial structure:

- **Percolation objects**:
  - It is no longer points, but **facets** ($(K-1)$-simplices) that merge.
- **Gabriel simplices**:
  - The simplices causing the merges (splitting simplices) are **Gabriel** simplices.
- **Computational domain**:
  - The search is restricted to the simplices of the **higher-order Delaunay mosaic** [11].

Comparison 1-MST vs. K-MST

	$K = 1$	$K \geq 2$
Vertices	Points	$(K-1)$-simplices
Merges	Edges	K-simplices
Condition	Gabriel	Gabriel
Domain	Delaunay	Order-K Delaunay

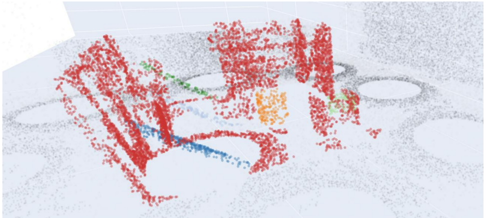
Summary: A Missing Link for Geometry Processing

- **Higher-order Delaunay mosaic**:
  - Contains the **entirety** of the hierarchical K-NN information.
  - Represents the optimal and sparse support for higher-order clustering.
- **Absence in existing libraries**:
  - **Observation**: To date, there is **no geometry processing library** (such as CGAL or Geogram) integrating this object.
  - **Consequence**: Need to develop complex *ad-hoc* code to construct and manipulate these cell complexes.

03
A Few Examples in 3D

Anomaly Detection on 3D Point Clouds (LiDAR)

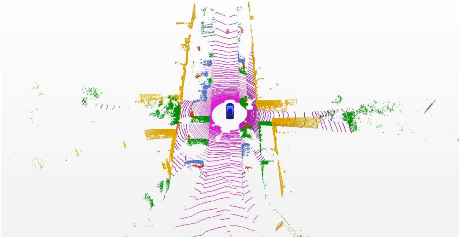
- **Geometric Prior**:
  - Comparison of the cluster hierarchy (produced by HGP-Clusterer) with the theoretical 3D model of the ship (© NAVAL GROUP).
- **Hierarchy Exploration**:
  - A cost function penalizes hybrid nodes containing both reference and anomaly points.
- **Result** (*Marie Benchmark*):
  - Perfect isolation of foreign objects (cubes, supports) attached to the structure.



Anomaly isolation result (reference model in red, anomalies in distinct colors).

Instance Segmentation on 3D/4D LiDAR

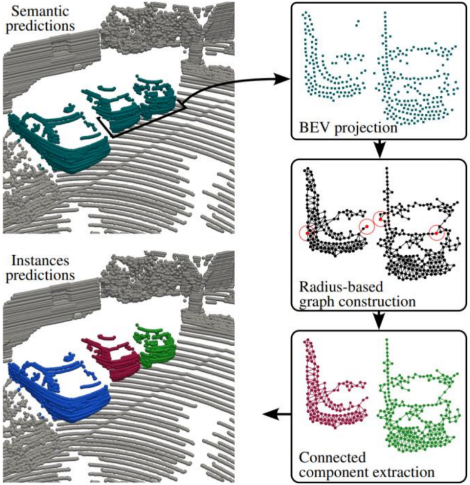
- **Industrial Perspectives**:
  - Collaborations and target applications with Greehill, Valeo, and Naval Group.
- **Automation**:
  - Navigation for autonomous vehicles and mobile robotics.
- **Monitoring & Inspection**:
  - Construction of urban or industrial digital twins.
  - Structural diagnosis and early anomaly detection.



Dense urban 3D LiDAR scene.

3D LiDAR Panoptic Segmentation: State of the Art

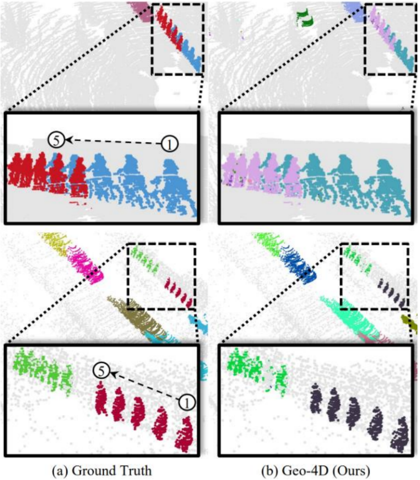
- **Standard Domain Approach**:
  - Semantic segmentation (Deep Learning) followed by spatial clustering (DBSCAN / HDBSCAN).
- **Major Limitations**:
  - Increased clustering difficulties in high-density areas or on complex geometries.
- **3D Reference**:
  - **Alpine** [13] (Sautier *et al.*, 2025):
  - *"Is clustering enough for LiDAR instance segmentation?"*



Visualization of instances detected by Alpine.

4D LiDAR Panoptic Segmentation: State of the Art

- **Temporal Tracking of Objects**:
  - Global association of semantic instances over time.
- **Standard Approach**:
  - Semantics + Clustering + Global association by Optimal Transport.
- **4D Reference**:
  - **Geo-4D** [10] (Oh *et al.*, 2025):
  - *"Training-Free Global Geometric Association for 4D LiDAR Panoptic Segmentation"*.



Temporal tracking of vehicle instances in Geo-4D.

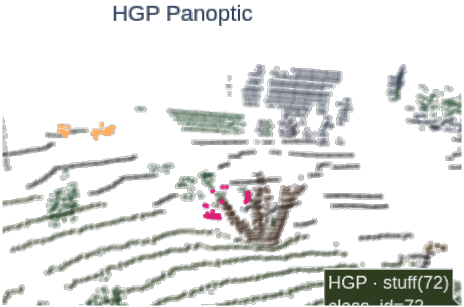
4D Instance Segmentation: Integrating Geometric Priors

- **Weak Geometric Priors**:
  - Taking into account the expected volume (weak prior) of the objects of interest (vehicles, pedestrians).
- **Selection in the Hierarchical Tree**:
  - Our algorithm identifies nodes that jointly optimize spatial density and the volume constraint.

Examples of geometric priors applied to instance volumes.

Temporal Instance Tracking (SemanticKITTI)

- **Robust Temporal Association**:
  - Continuous tracking of instances across successive frames.
- **Trajectory Reconstruction**:
  - Minimization of the global geometric error from one frame to the next.

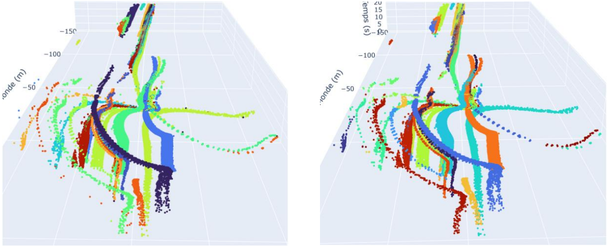


Estimated vehicle trajectories over time.

4D Instance Segmentation Results

- **Our Algorithm**:
  - Higher-order hierarchical clustering combined with unbalanced Optimal Transport (unbalanced OT).
- **Training-Free**:
  - Robust results completely free of deep learning for the tracking and clustering part.

Ground Truth (GT) **Our Algorithm**



Visual comparison of reconstructed trajectories on SemanticKITTI.

References I

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